

Gastric Secretion

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LEARNING OBJECTIVE

At the end of lecture, student should be able to

- Describe the functions of stomach
- List the secretions of the stomach, their cells of origin & their function
- Discuss the various factors that stimulate & inhibit the secretory activity of the parietal cell
- Discuss the long & short neural reflexes that influence HCl secretion
- Discuss the hormonal influences on HCl secretion
- Illustrate the different phases of gastric secretions
- Discuss the potentially damaging effects of HCl on gastric mucosa
- Composition and function of gastric secretions.
- Regulation of secretion mechanism of stimulating gastric secretion.

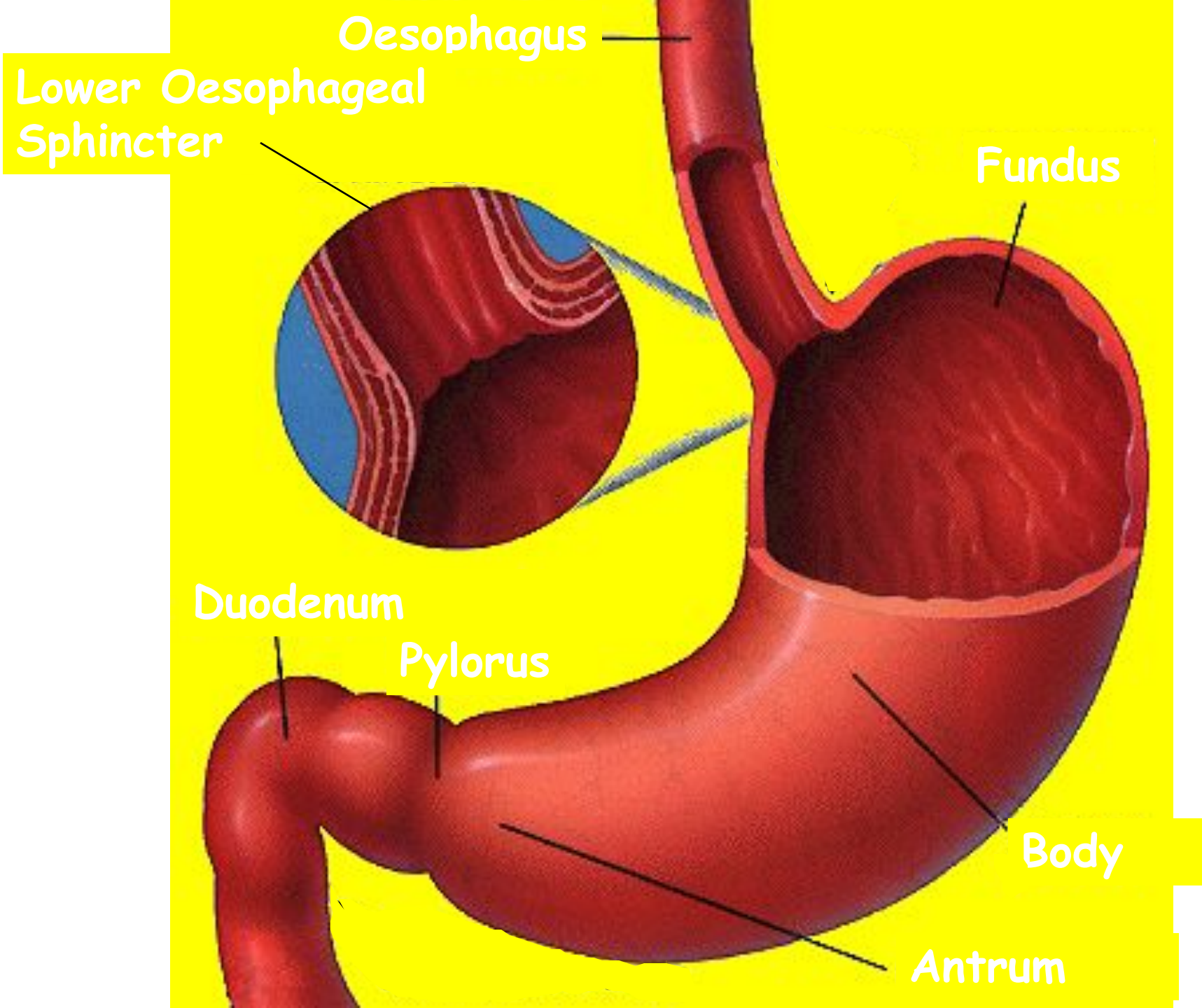
Functions of Stomach

- **Temporary store of ingested material**
- **Dissolve food particles and initiate digestive process**
- **Control delivery of contents to small intestine**
- **Sterilize ingested material**
- **Produce intrinsic factor (Vitamin B₁₂ absorption)**

Gastric Secretions (Gastrointestinal Physiology)

Youtube Link:

<https://youtu.be/QpUBsxJYmXg>



Types of Gastric glands

1. Oxyntic glands (Gastric glands)

Parietal (Oxyntic) cells; HCl & Intrinsic factor

Peptic (Chief) cells; Pepsinogen

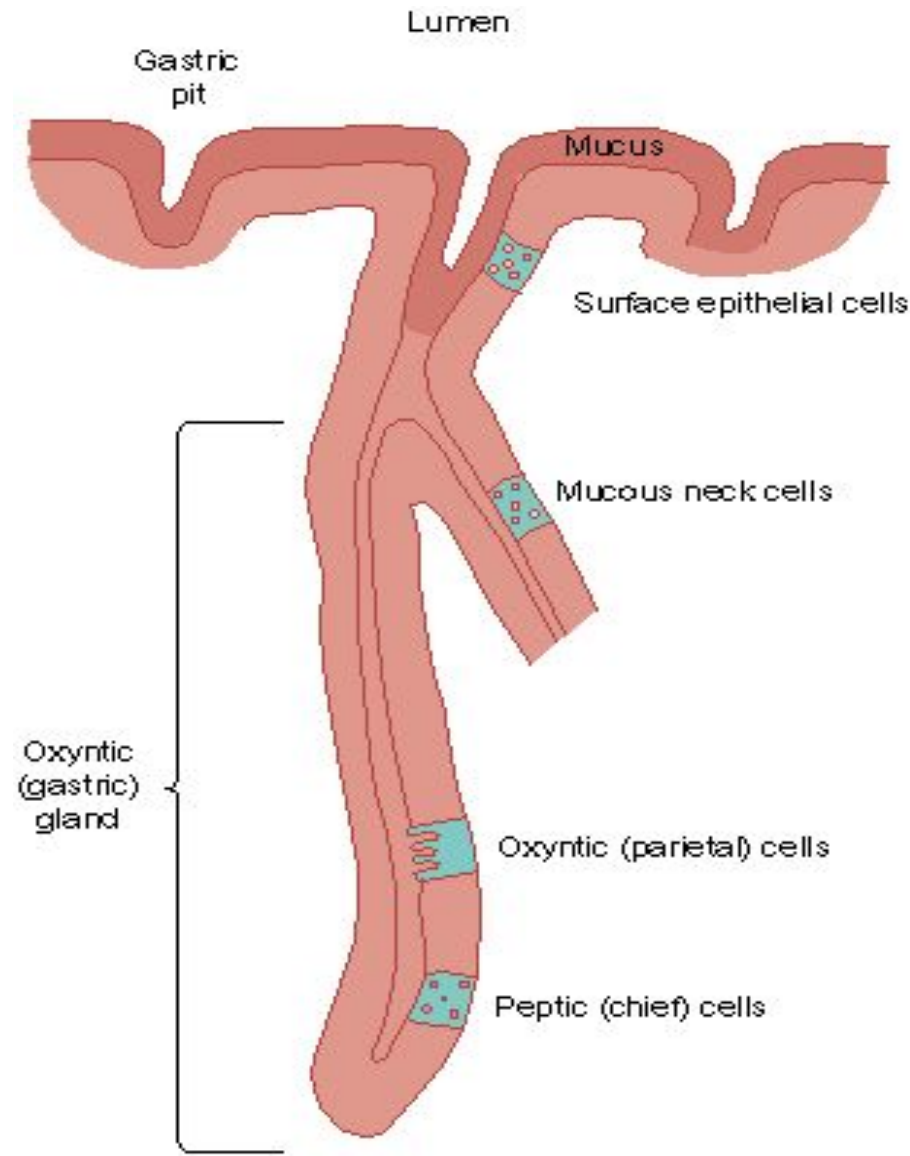
Mucous neck cell; secretes mucus

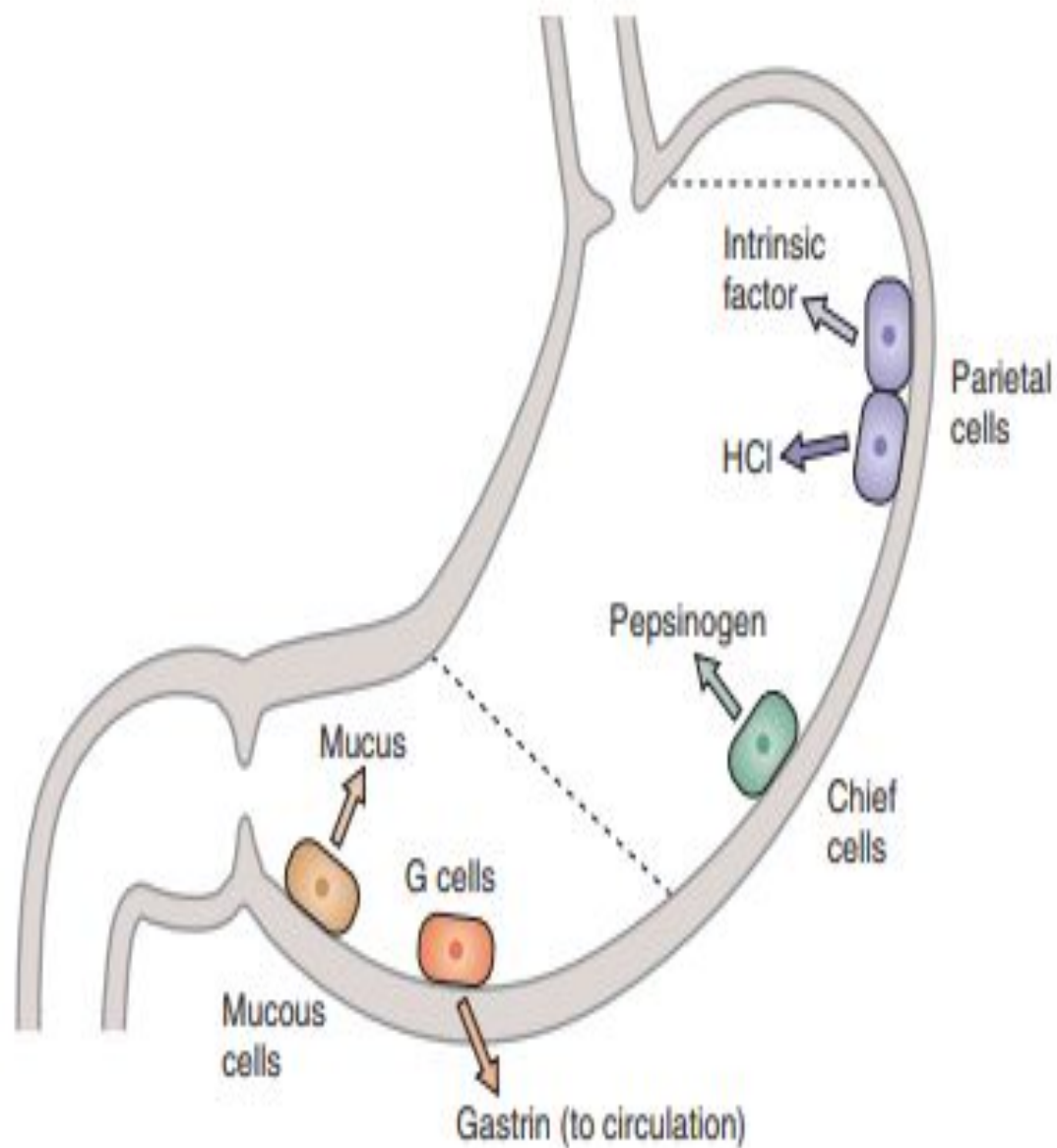
2. Pyloric glands

Mucous neck cells

Peptic cells

G cells secrete Gastrin

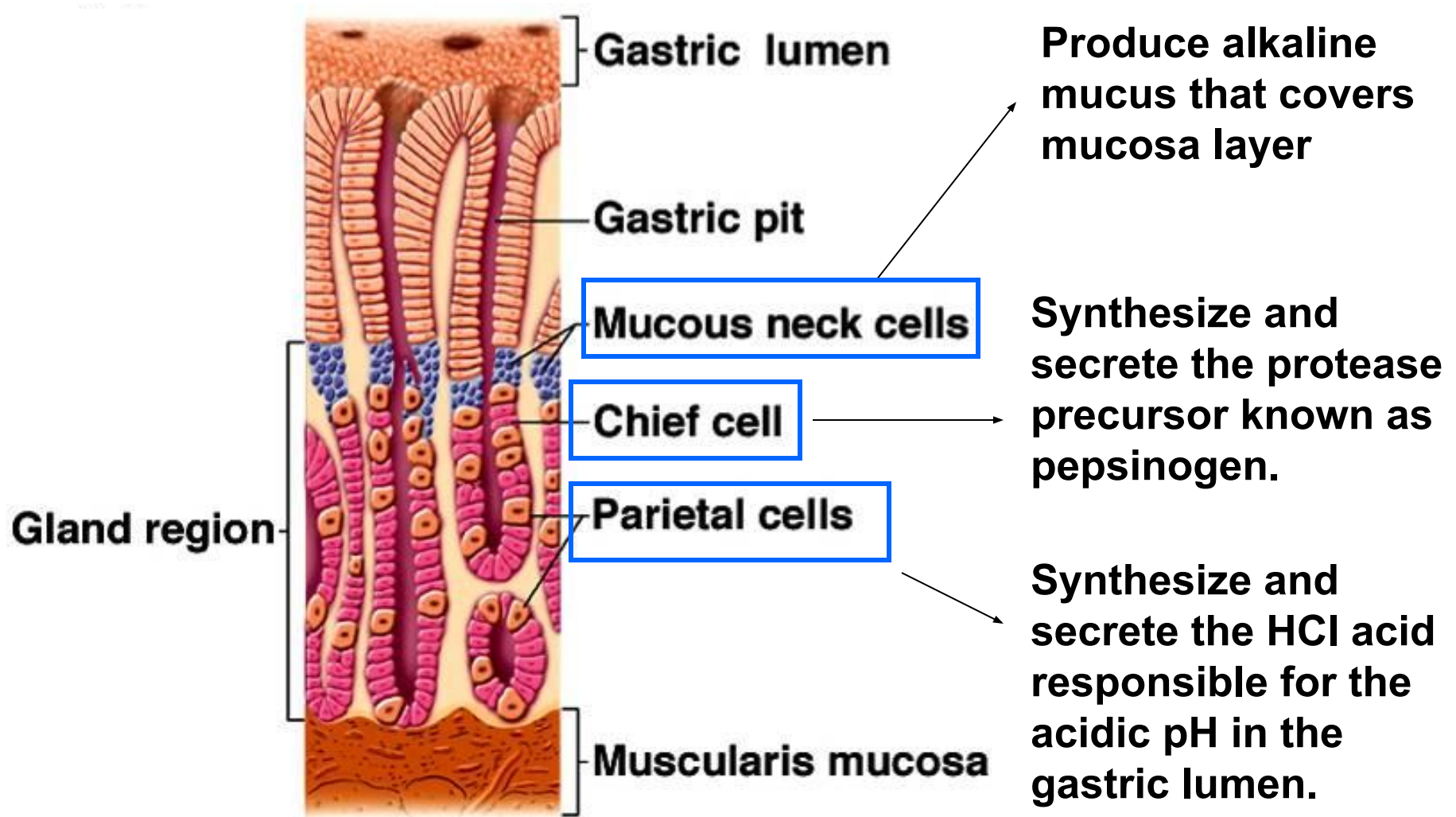




Cell Type	Location	Secretion
Parietal cells	Body	HCl Intrinsic factor
Chief cells	Body	Pepsinogen
G cells	Antrum	Gastrin
Mucous cells	Antrum	Mucus Pepsinogen

Fig. 8.15 Secretory products of various gastric cells.

Exocrine gland cells of gastric pits



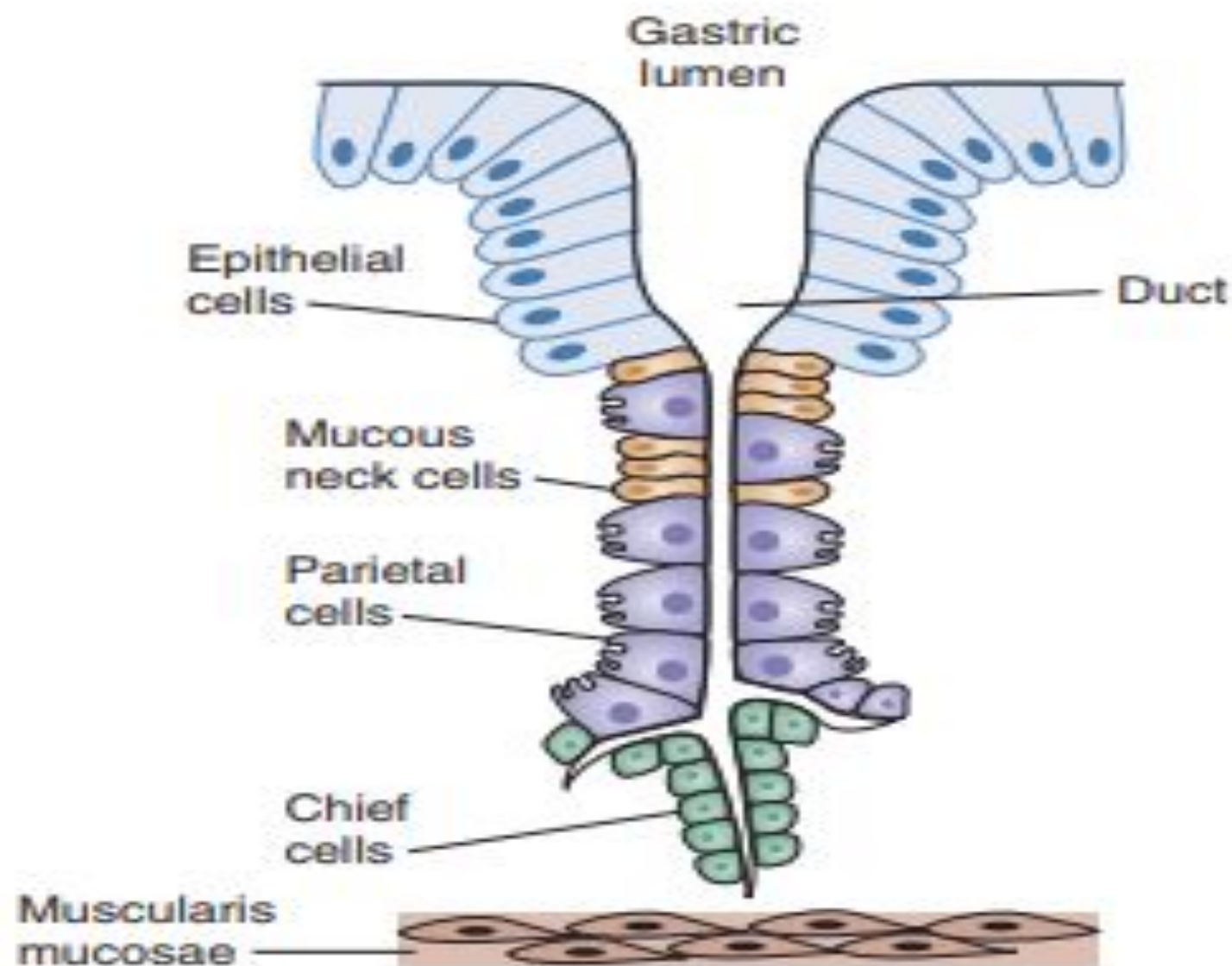


Fig. 8.16 Structure of a gastric oxyntic gland showing the various cell types lining the gland. The ducts open into pits on the surface of the gastric mucosa.

Composition and function of gastric secretions

1. HCl

- converts pepsinogen to pepsin for chemical digestion
- provides optimal pH environment for pepsin
- destroys some bacteria
- stimulates the small intestinal mucosa to release secretin and CCK
- promotes the absorption of Ca^{2+} and Fe^{2+} in small intestine



Composition and function of gastric secretions

2. Pepsinogen (precursor of pepsin)

digestion of proteins

3. Mucus

forms a protective barrier: Mucus-bicarbonate barrier

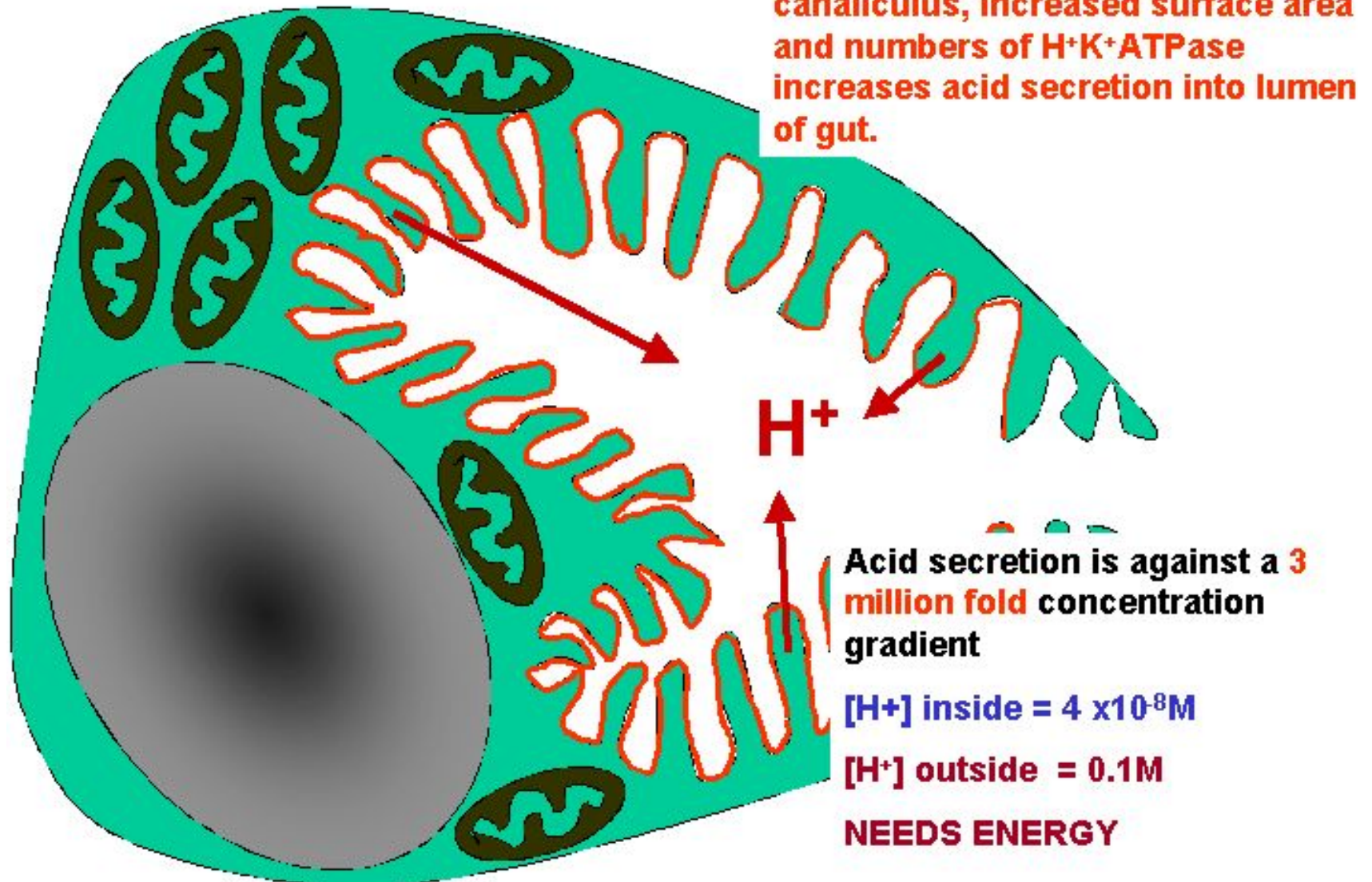
4. Intrinsic factor

combines with vitamin B₁₂ to make it absorbable



HCl secretion

Parietal cell - secreting



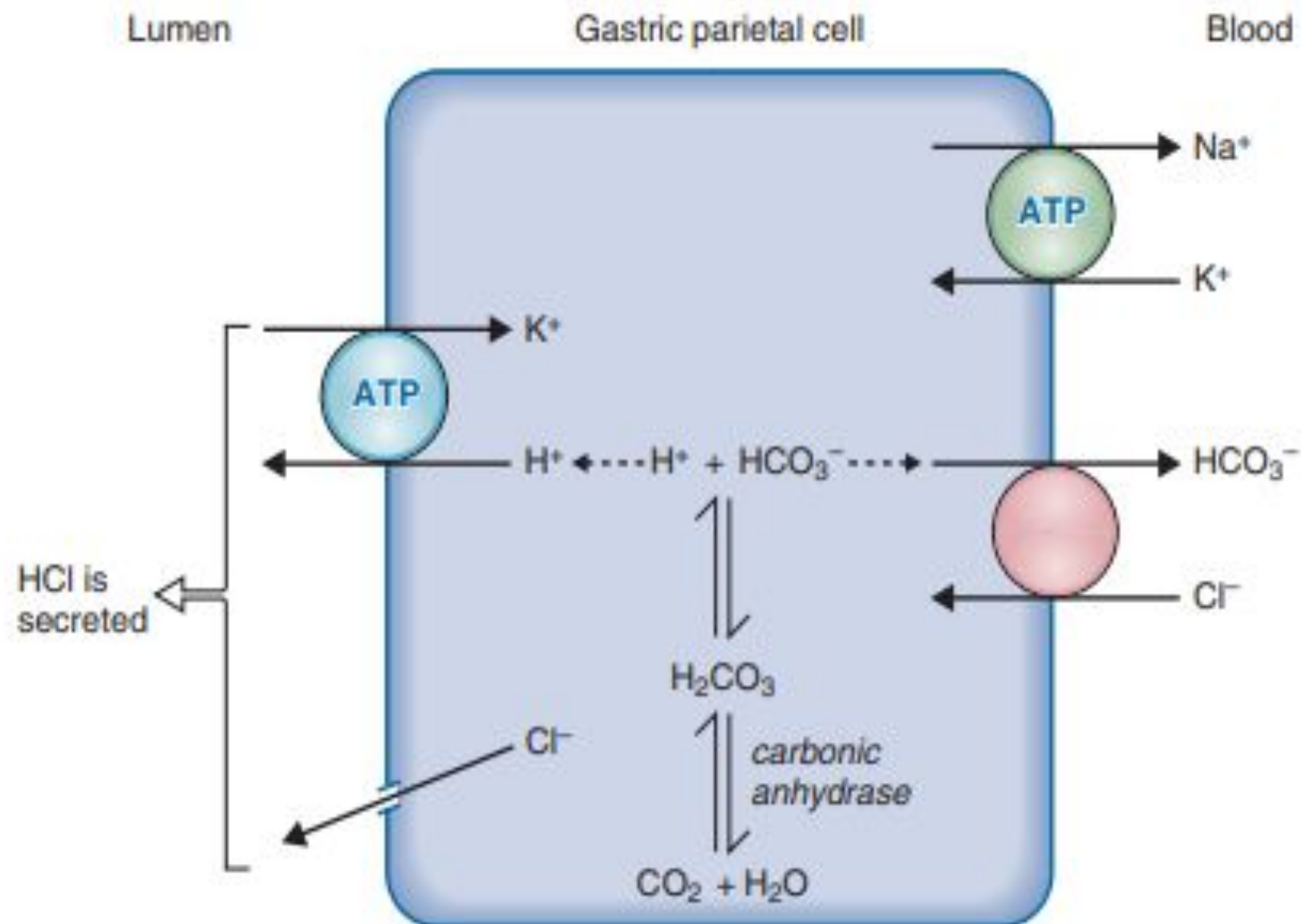


Fig. 8.17 Mechanism of HCl secretion by gastric parietal cells. ATP, Adenosine triphosphate.

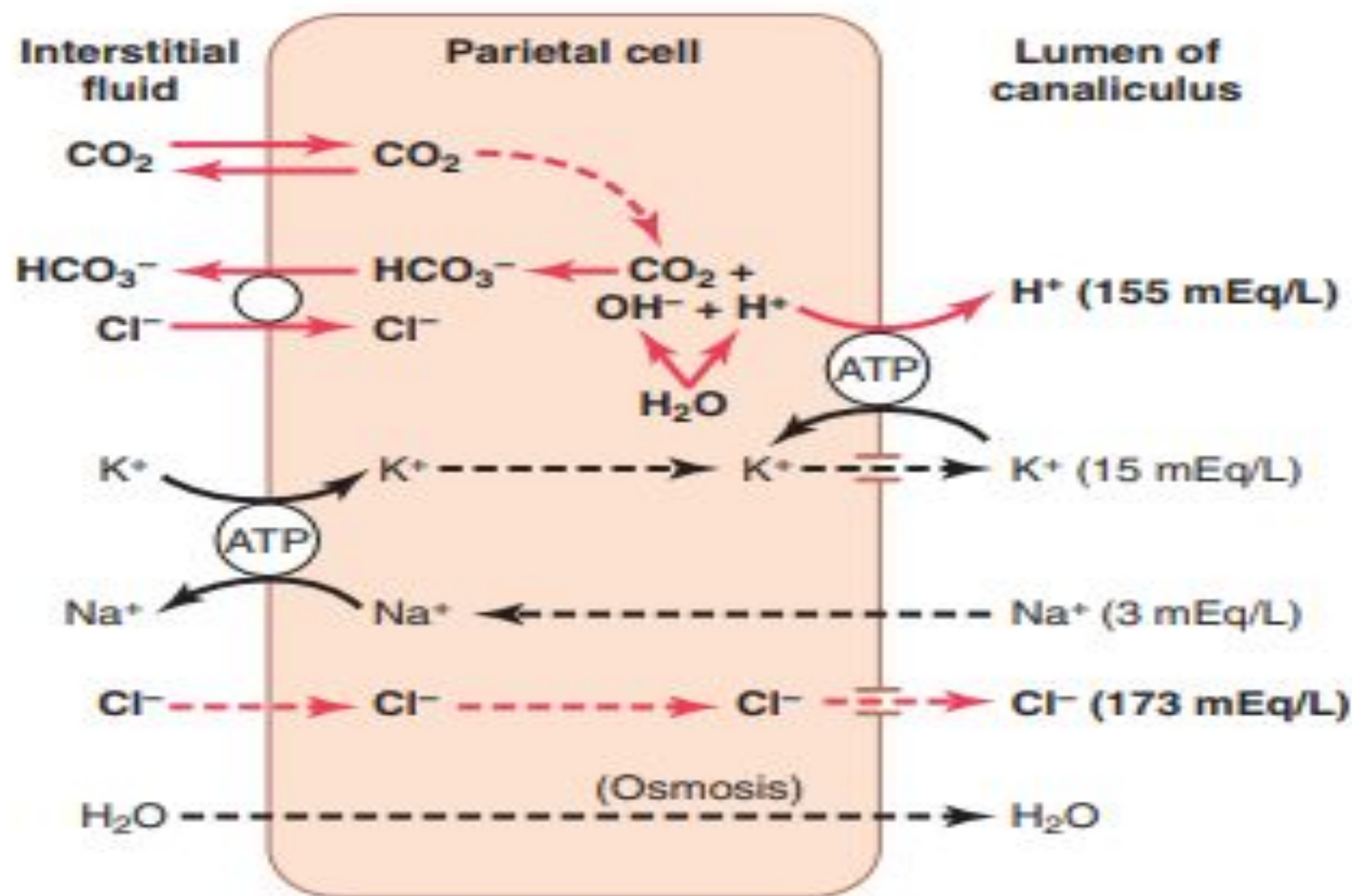


Figure 65-6. Postulated mechanism for secretion of hydrochloric acid. (The points labeled "ATP" indicate active pumps, and the *dashed lines* represent free diffusion and osmosis.)

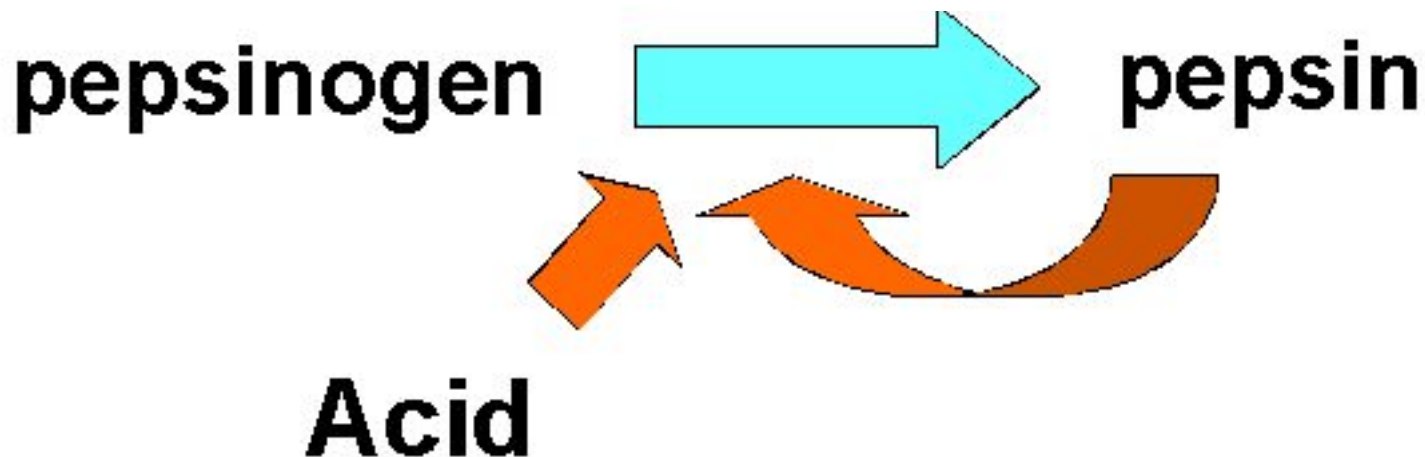
Gastric secretions

- **Pepsinogen**

Inactive precursor of pepsin
which initiates protein digestion

Is not necessary for complete
digestion of dietary protein –
pancreatic enzymes are sufficient

Active only when the $\text{pH} < 3.5$

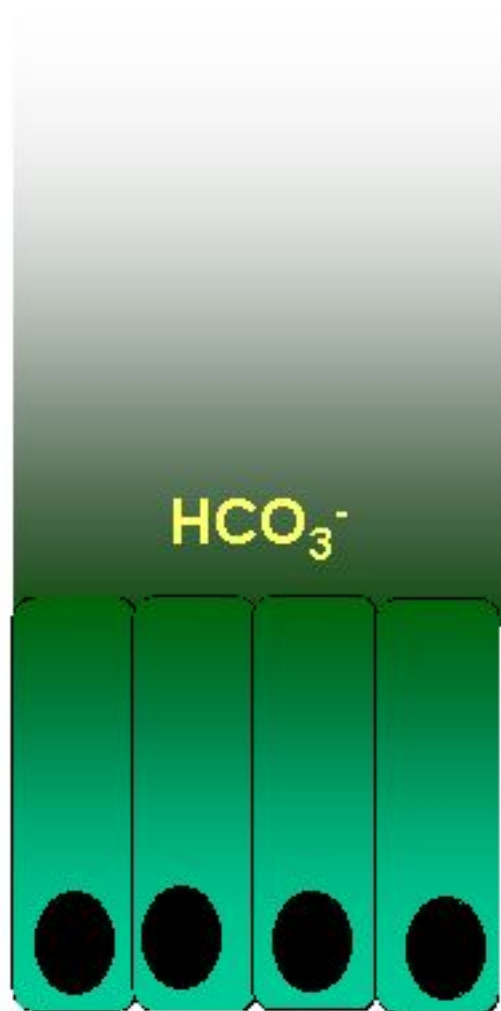


Gastric secretions

- **Mucus**

pH<2

pH7



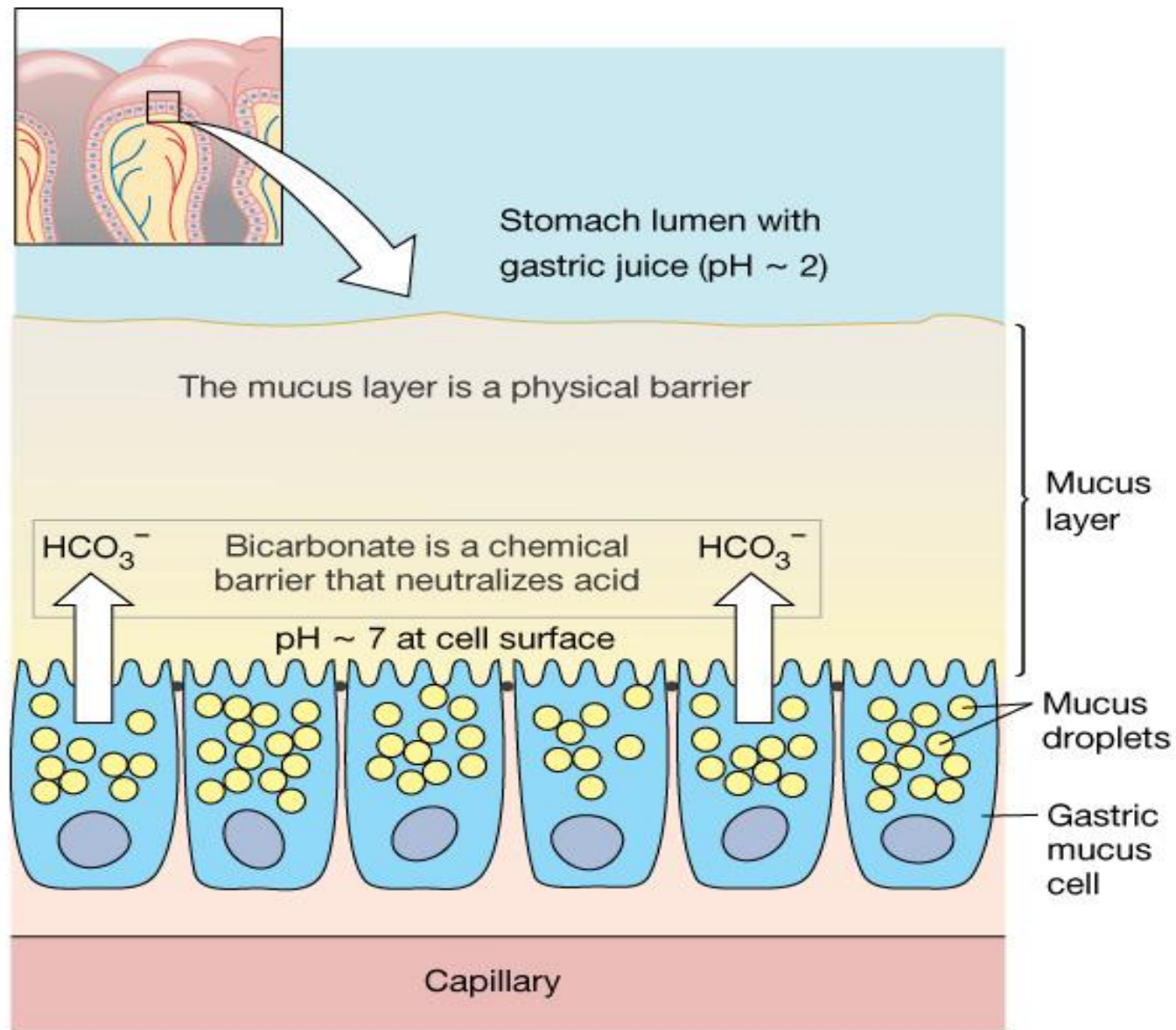
Physical/chemical barrier
to attack by gastric juice

Stimulated by:

- Ach
- Mechanical Stim
- Chemicals (ethanol)

If breached e.g.
hypersecretion of acid -
ulceration

Gastric Mucus-Bicarbonate Barrier



Gastric Mucus-bicarbonate barrier

The insoluble mucus and bicarbonate construct a barrier

- prevent hydrogen ions from diffusing to the mucosal layer
- protect the stomach mucosa from injury by hydrochloric acid and pepsin,

Intrinsic Factor

Only gastric secretion that is Essential for health

Secreted from parietal cells in humans, chief cells in other species

Forms a complex with vitamin B₁₂ in the gut

The complex is resistant to digestion and therefore enables absorption of vitamin B₁₂

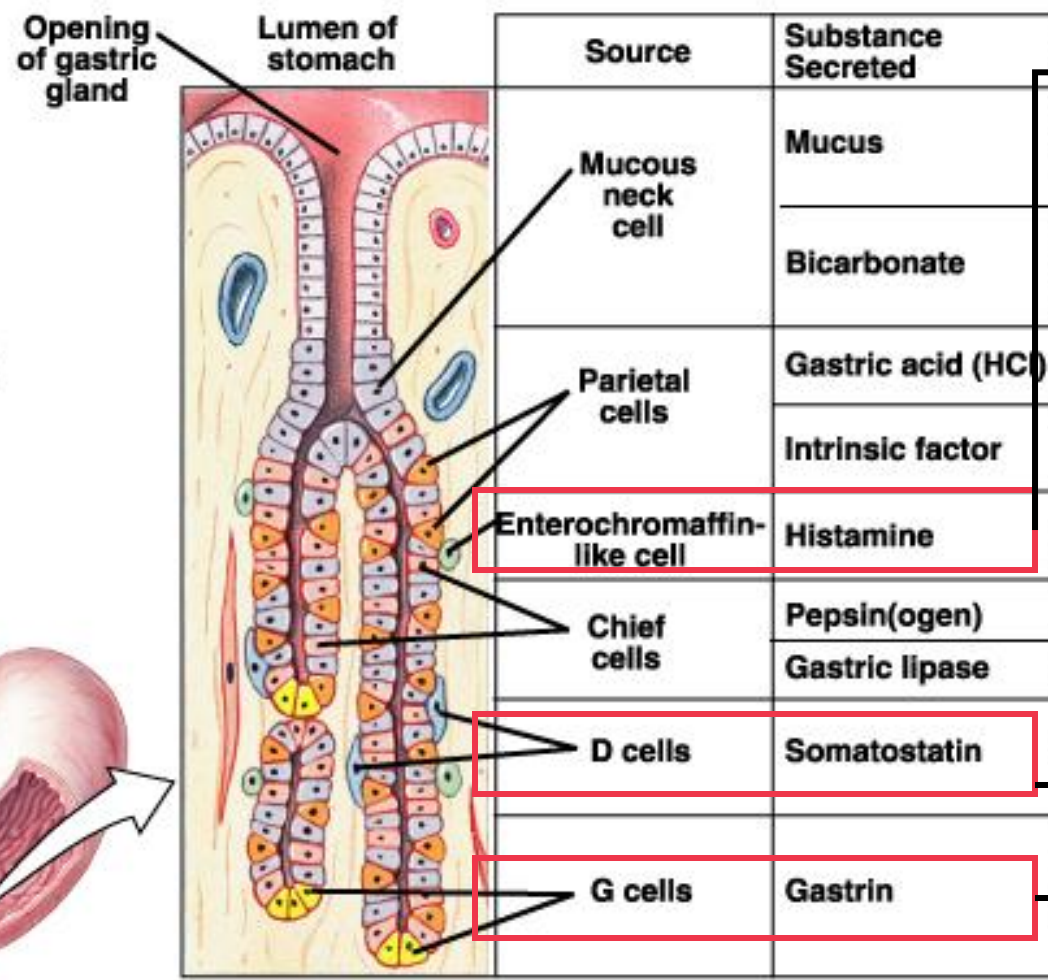
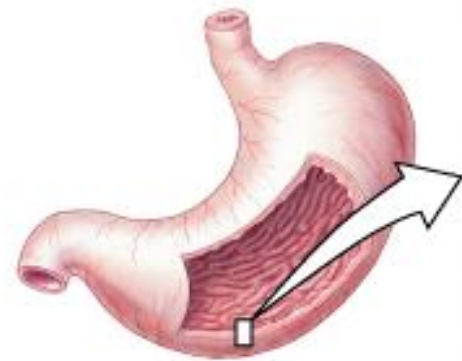
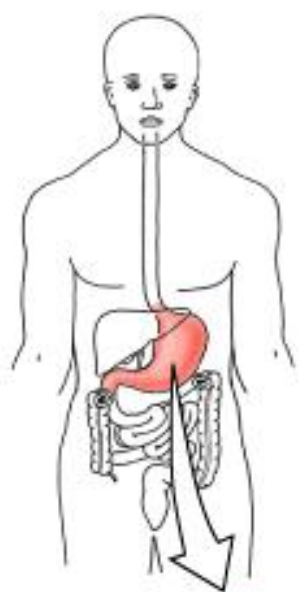
Lack of intrinsic factor causes Vit B₁₂ deficiency (pernicious anaemia) – as all the Vit B₁₂ is digested and therefore can not be absorbed

Regulation of Secretion

Control of Gastric Acid Secretion

Gastric acid secretion is controlled by three mechanisms:

- Neurocrine (vagus/local reflexes)
- Endocrine (gastrin)
- Paracrine (histamine)



Stimulates acid secretion

Inhibits

- acid secretion
- gastrin and pepsin release
- pancreatic exocrine secretions

Stimulates acid secretion

Regulation of Gastric Secretions

The important stimulatory signals

Autonomic nerves

- Release ACh
- Stimulates smooth muscle contraction
- Also stimulates Chief , Parietal , ELC and G cells

Gastrin

- Stimulates Chief , Parietal , ELC cells

Histamin

- Stimulates Parietal cells

Protein products such as peptides, A.A's

- Stimulates G-cells

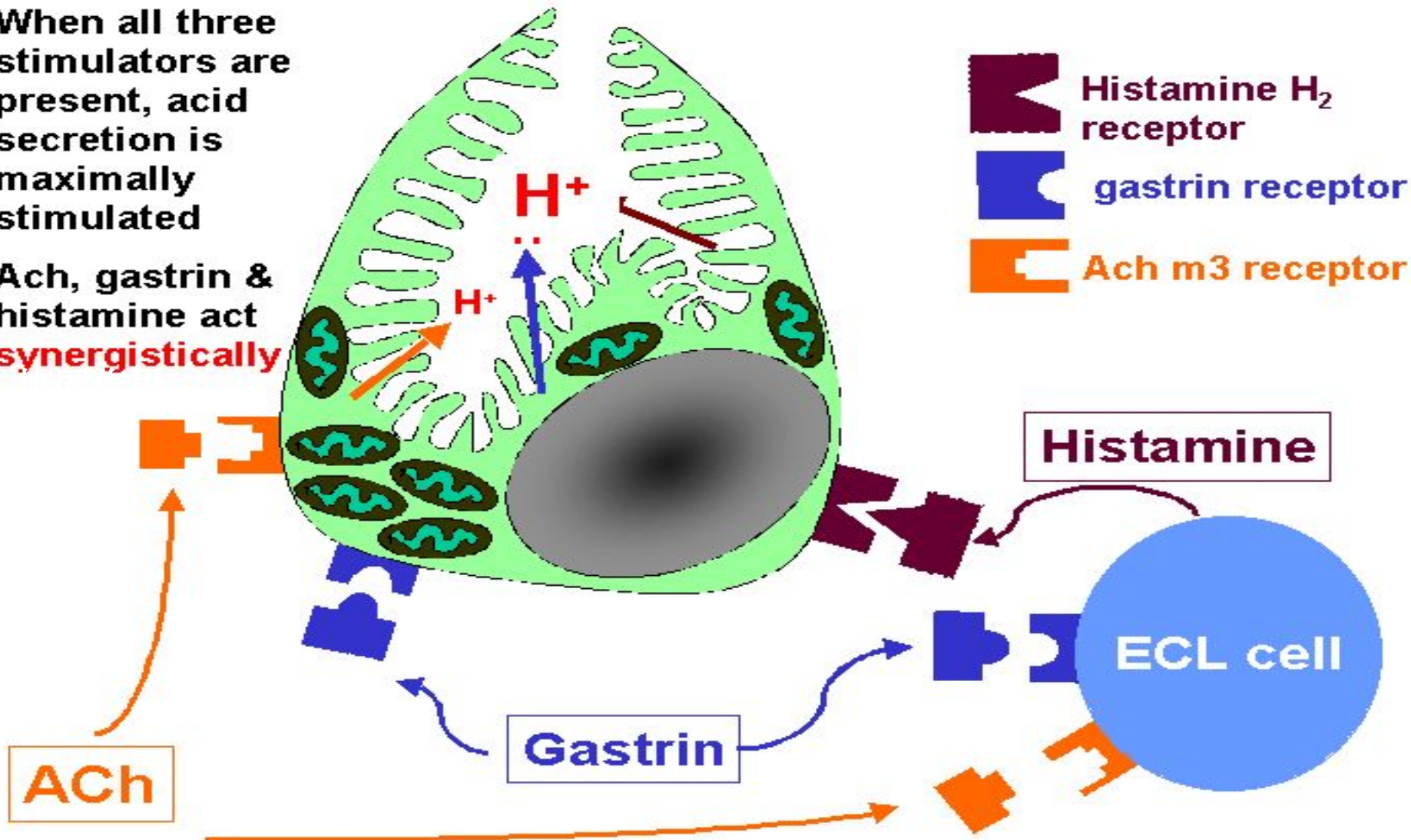
Acids

- Stimulate D cells

Endogenous substances regulating gastric secretion

When all three stimulators are present, acid secretion is maximally stimulated

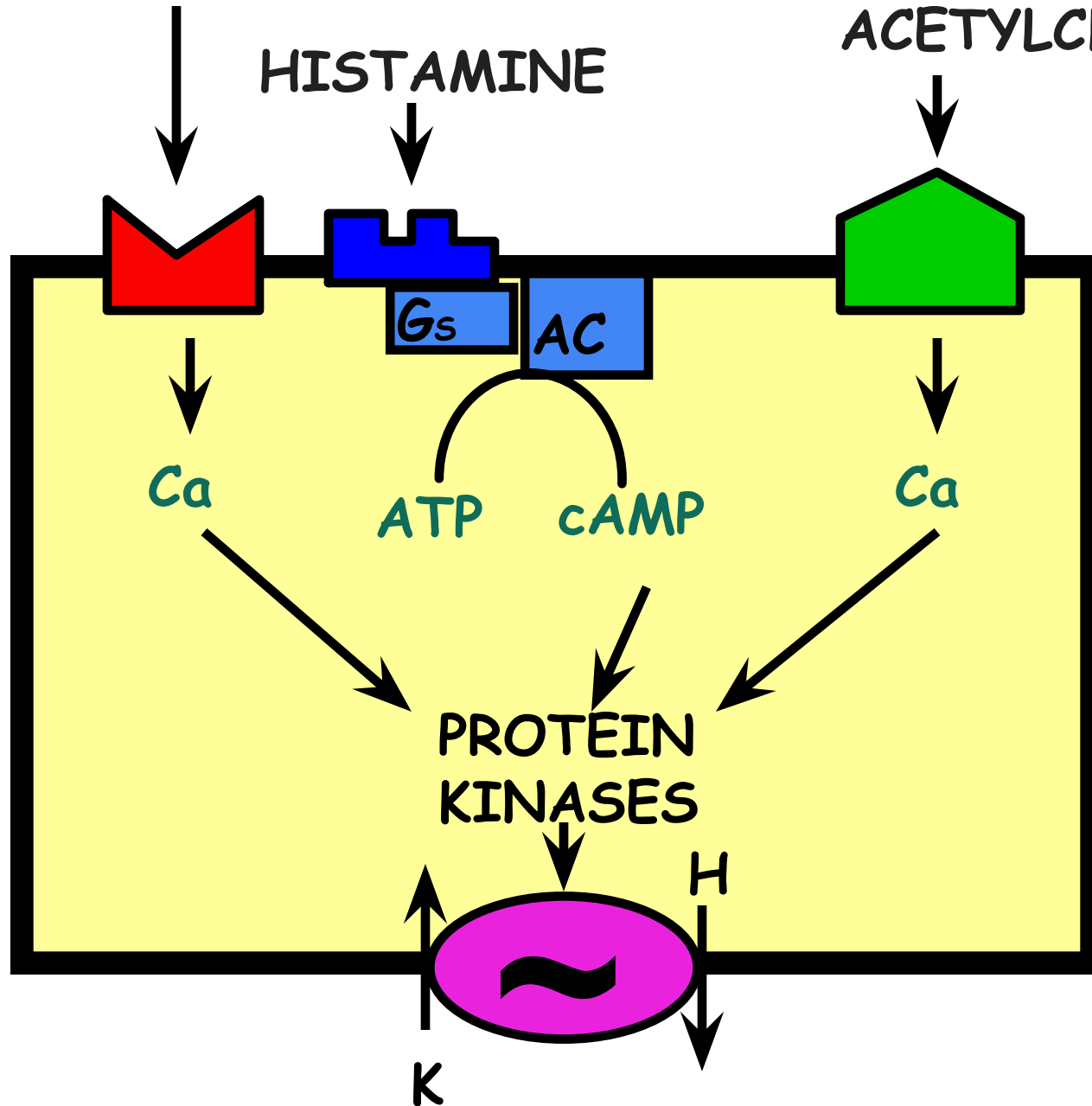
Ach, gastrin & histamine act synergistically



GASTRIN

HISTAMINE

ACETYLCHOLINE



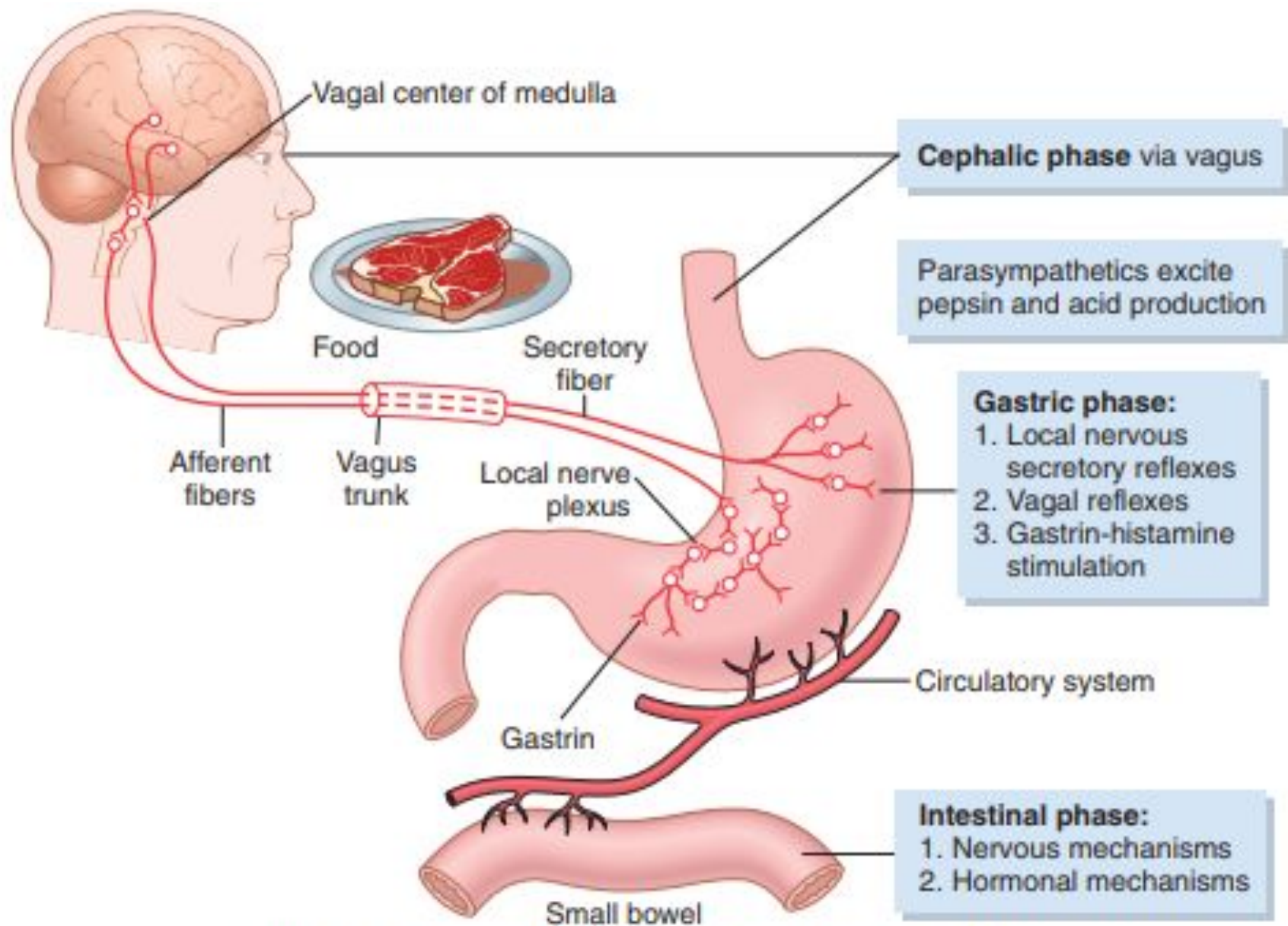


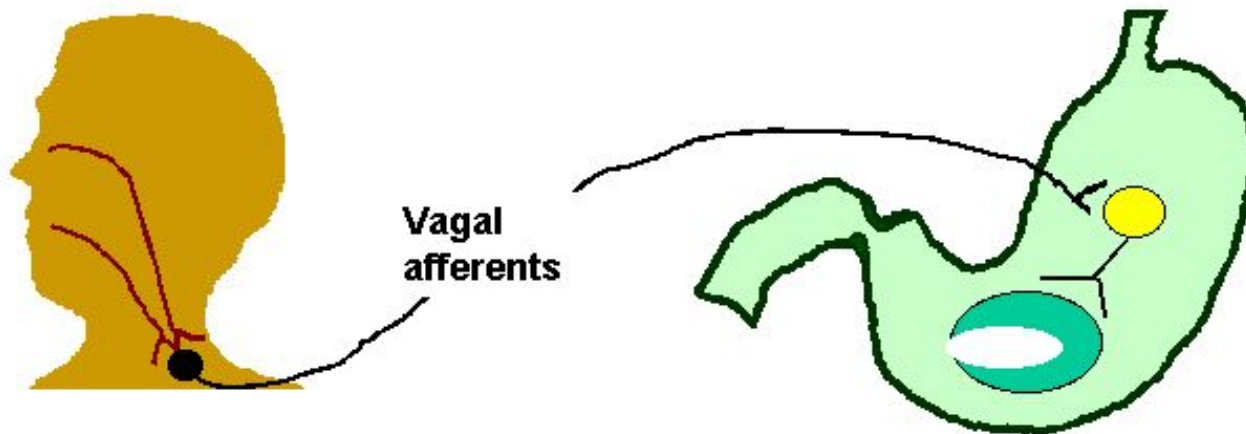
Figure 65-7. Phases of gastric secretion and their regulation.

1. Cephalic phase

- Occurs before food enters the stomach;
- initiated by smell, taste, sight Impulses from olfactory, chemical and other receptors activate the vagal nuclei in the medulla (via Hypothalamus)
- This triggers motor impulses to travel via the vagus nerve to the parasympathetic enteric ganglia
- Enteric ganglia in turn stimulate stomach glands

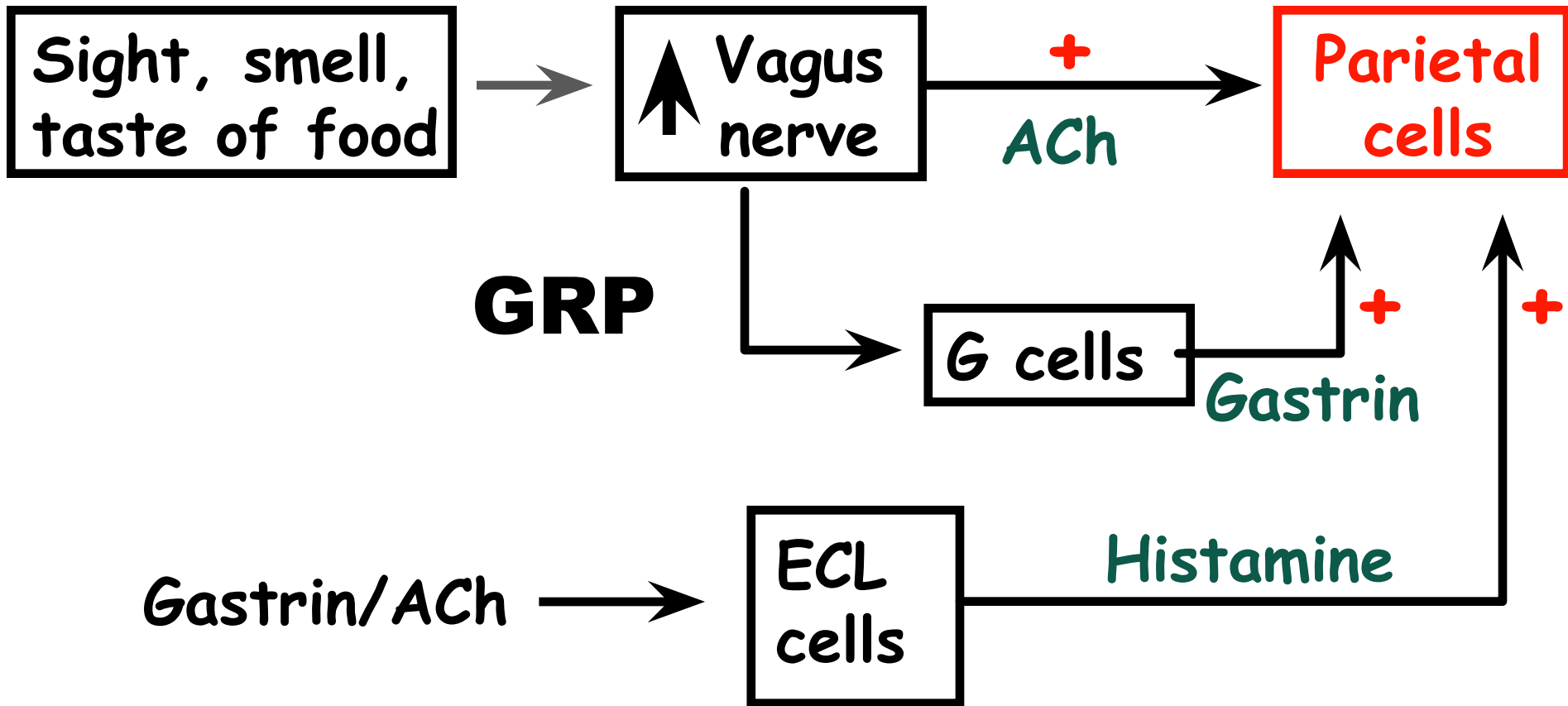
Cephalic Phase

- Unconditioned and conditioned reflex
- Only occurs when we want food
- depression dampens this reflex
- Account for 30% total volume of secretion
- Large amount of HCl and pepsinogen, high digestive ability

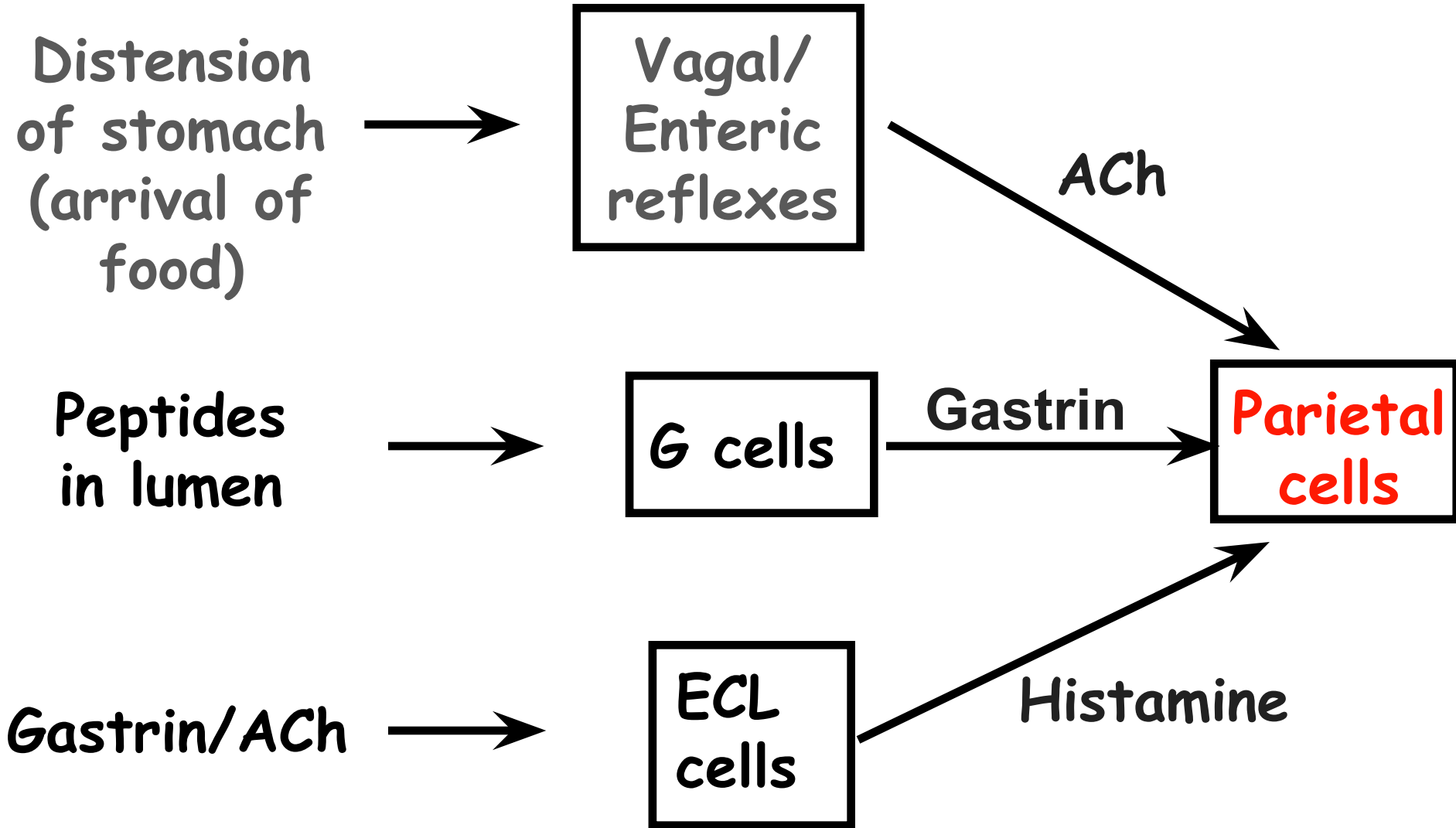


Mechanisms Stimulating Gastric Acid Secretion in Cephalic Phase

Cephalic Phase



Gastric Phase



- Starts when food reaches the stomach
- Provides 2/3 of the juice released
- Accounts for 60% of gastric secretions
- There are two parts (neural and chemical) to this phase
 - Neural part
- Activated by stretch receptors
- Initiates both local neural reflexes as well as the longer vago-vagal reflex
- Both reflexes result in release of ACh at stomach synapses which stimulates secretory cells
- This branch is inhibited by Sympathetic action (emotional upset)

Chemical part

- An increase in pH (thus, 'less' acidity), presence of peptides, caffeine activate the G-cells
- This results in Gastrin being released
- Gastrin acts on Parietal cells that start secreting HCl
- Gastrin also stimulates Histamine release, which in turn stimulates Parietal cells
- The increase in HCl promotes pepsin production and protein degradation

Chemical part

The release of Gastrin is partly regulated by acidity

- Increased acidity inhibits the G-cells
- Increased presence of proteins in a meal tends to buffer proton
- This in turn keeps the pH from becoming too acidic and allows more gastrin to be released

Intestinal Phase

- Account for about 10% of secretion
- Primarily hormonal – denervated stomach will be stimulated to secrete acid by protein in duodenum
- Hormone still unknown
- Very small number of G-cells in duodenum also release gastrin in response to amino acids

Regulation of Gastric Secretions occurs via 3 phases

TABLE 15-5 Control of HCL Secretion during a Meal		
STIMULI	PATHWAYS	RESULT
<i>Cephalic phase</i> Sight Smell Taste Chewing	Parasympathetic nerves to enteric nervous system	↑HCl secretion
<i>Gastric contents (gastric phase)</i> Distension ↑Peptides ↓H ⁺ concentration	Long and short neural reflexes, and direct stimulation of gastrin secretion	↑HCl secretion
<i>Intestinal contents (intestinal phase)</i> Distension ↑H ⁺ concentration ↑Osmolarity ↑Nutrient concentrations	Long and short neural reflexes; secretin, CCK, and other duodenal hormones	↓HCl secretion

Mechanisms Inhibiting Gastric Acid Secretion

Cephalic Phase

Stopping eating $\longrightarrow$ Vagal activity

Gastric Phase

$\downarrow$ pH ($\uparrow$ [HCl]) $\longrightarrow$ Gastrin

Intestinal Phase

Acid in duodenum $\longrightarrow$ Enterogastric (splanchnic) reflex
 $\downarrow$ Secretin release

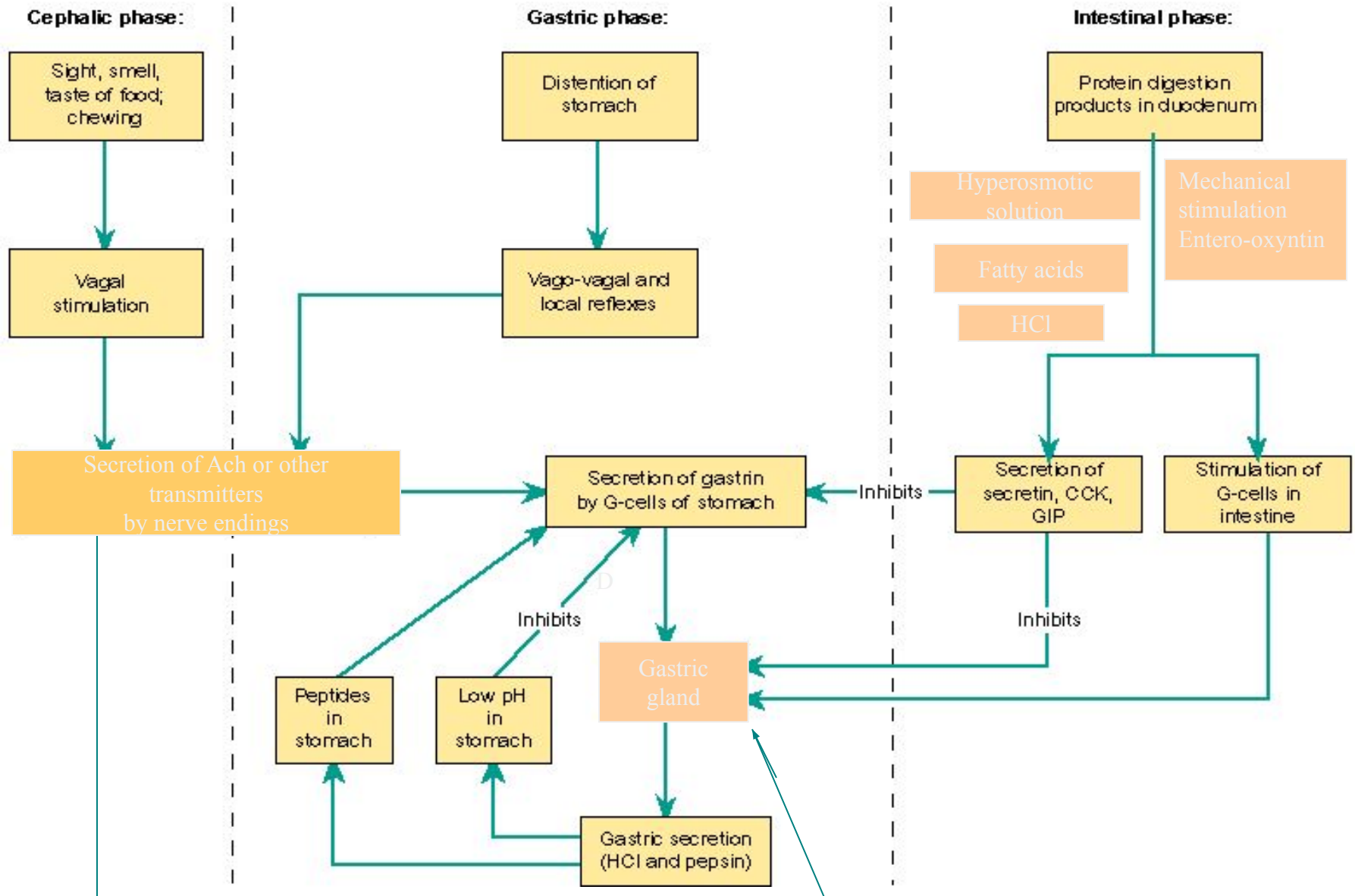
$\left. \begin{array}{l} \text{Enterogastric (splanchnic) reflex} \\ \text{Secretin release} \end{array} \right\} \longrightarrow \begin{array}{l} \text{Gastrin secr}^n \\ \text{Gastrin stim}^n \\ \text{of parietal} \\ \text{cells} \end{array}$

Fat in duodenum $\longrightarrow$ GIP release $\longrightarrow$ Gastrin secrⁿ
 $\downarrow$ Parietal HCl secrⁿ

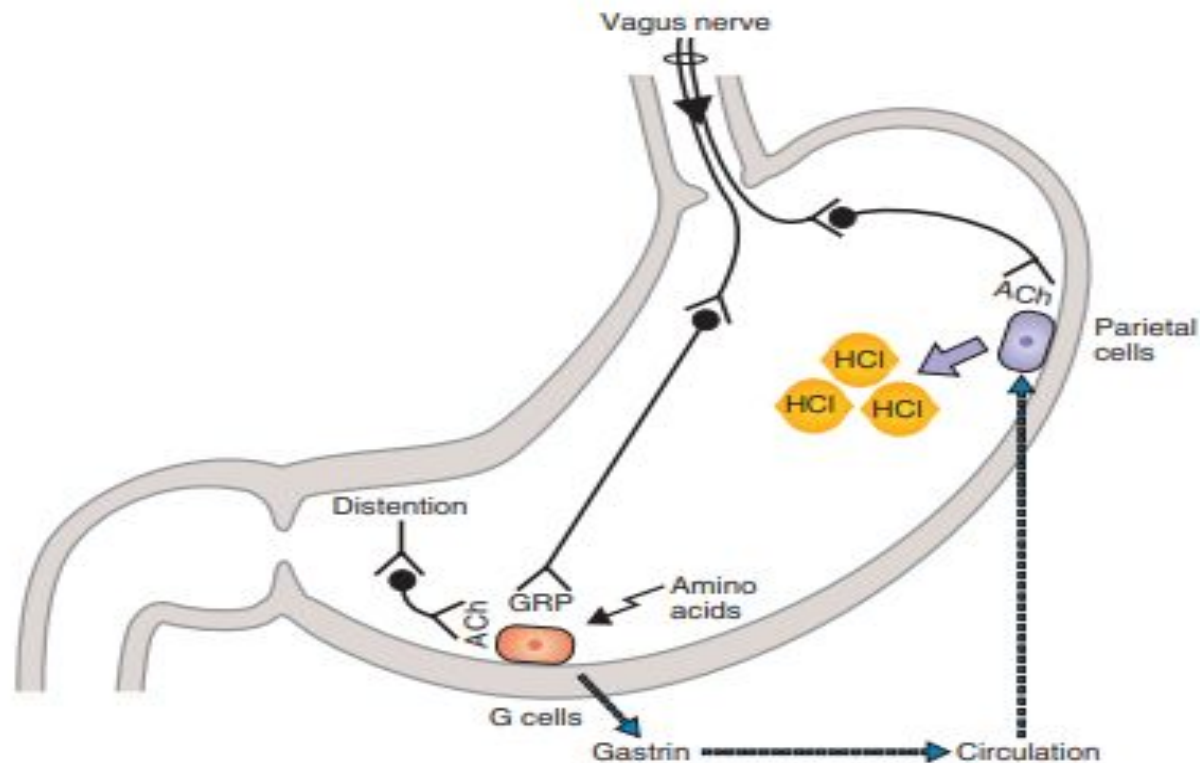
Enterogastrones

- Hormones released from gland cells in duodenal mucosa - secretin, cholecystokinin (CCK), GIP
- Released in response to acid, hypertonic solutions, fatty acids or monoglycerides in duodenum
- Act collectively to prevent further acid build up in duodenum
- Two strategies:
 - inhibit gastric acid secretion
 - reduce gastric emptying (inhibit motility/contract pyloric sphincter)

Regulation of gastric secretion



REGULATION OF HCl SECRETION



Phase	% of HCl Secretion	Stimuli	Mechanisms
Cephalic	30%	Smell, taste, conditioning	Vagus → parietal cell Vagus → gastrin → parietal cell
Gastric	60%	Distention	Vagus → parietal cell Vagus → gastrin → parietal cell
		Distention of antrum	Local reflex → gastrin → parietal cell
		Amino acids, small peptides	Gastrin → parietal cell

Fig. 8.19 Regulation of HCl secretion during cephalic and gastric phases. ACh, Acetylcholine; GRP, gastrin-releasing peptide (bombesin).